

Process Structure

⇒ Oracle takes advantage of various types of processes:

⇒ There are three types of processes:

1. User Process
2. Server Process
3. Background processes

User Process:

⇒ It is started when a database user or a batch process connects to Oracle Database.

⇒ User processes are generated at the client end.

In other words:

⇒ A program that requests interaction with the Oracle server.

⇒ Must first establish a connection.

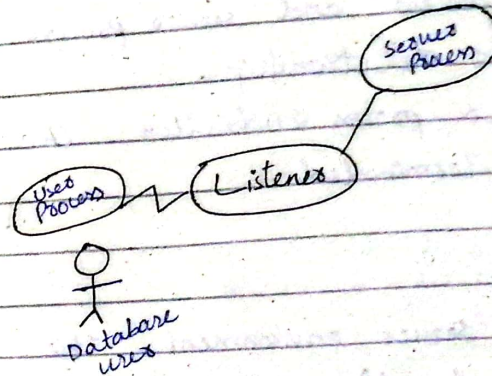
⇒ Does not interact directly with Oracle server.

⇒ Connection could be via Oracle provided tools like SQL*Plus, SQL Developer or a third-party tool like TOAD etc.

⇒ Each user process is linked to a dedicated server process through which its tasks are performed.

⇒ A user request and response is handled by the server process.

⇒ User process provides the means of communication among application running at the user end and the Oracle instance



2. Server Process:

⇒ Connects to the Oracle instance.

⇒ Started when a user establishes a connection.

⇒ A program that directly interacts with the server.

⇒ Dedicated process for a user process

⇒ The server process communicates on behalf of the user-process with the Oracle database server.

⇒ A server process can be either a dedicated server process or a shared server process.

Dedicated server:

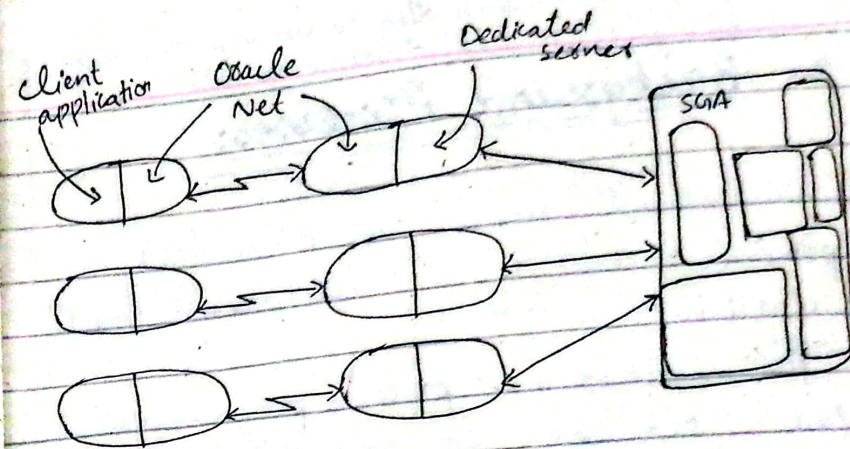
⇒ In Dedicated ^{uses} process and server process has one-to-one relationship.

⇒ when user process ends then the server process terminated.

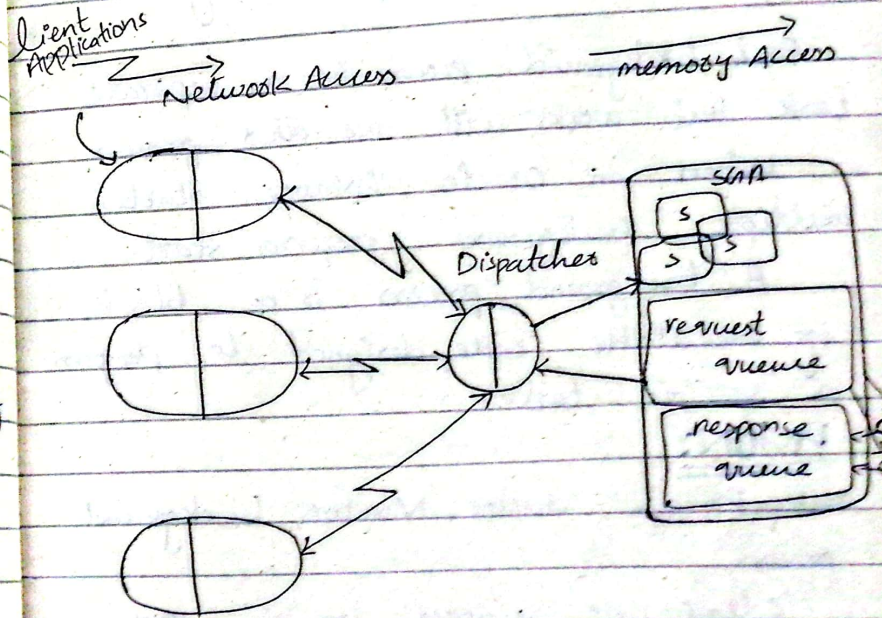
Shared server:

⇒ In shared server environment, the server process handles the request of several user processes.

⇒ The server process communicates with the Oracle server using the Oracle program interface.



Dedicated server connection



Shared server connection

Functions

- Interact with the OS.
- Interact with each other to manage memory structures.
- Manage the actual database on disk.

3. Background Processes:

⇒ Background processes are additional processes used by a multi-process Oracle database.

⇒ The background processes perform maintenance tasks required to operate the database and to maximize performance for multiple users.

⇒ Each background process has a separate task, but works with the other processes.

⇒ When an Oracle instance starts, multiple background processes start.

⇒ A background process is a block of executable code designed to perform a specific task.

PMON:

⇒ Stands for Process Monitor - Monitors background processes.

⇒ Clean up resources if a user connection is dropped or fails.

⇒ Removes locks on affected rows.

⇒ Rolls back transactions in progress.

⇒ Marks transaction blocks as available in

cache buffers.

⇒ Updates status of instance for incoming connection requests.

2. SMON:

⇒ Stands for System Monitor Process.

⇒ Performs crash recovery by applying entries in online redo log files to data files in case of power outage or CPU failure.

⇒ Performs system level cleanup activities.

⇒ Recovers all committed transactions.

⇒ Opens the database to system users to logon.

⇒ Rollback the uncommitted transactions.

⇒ Limited space management.

⇒ Purges (completely removes) temporary segments during system restart.

⇒ Coalesces (merge or combine) free space in tablespaces regularly if dictionary managed.

3. DBWn:

⇒ Stands for Database Writer.

⇒ Writes modified / dirty blocks from the database buffer cache to the datafiles.

⇒ Writes oldest, least active block first using LRU algorithm.
⇒ Up to 80 DBWn processes can be started.

→ DBWD - DBW9

→ DBWA - DBW1

⇒ Writes when:

- CKPT occurs
- Dirty buffers reach the threshold
- Timeout occurs
- There are no free buffers
- TBS is offline
- TBS being backup mode

⇒ The number of DBWn processes is controlled by the DB-WRITER-PROCESS Parameter.

4) LGWR:

⇒ Stands for Log writer.
⇒ Manages redo log buffers.
⇒ Writes redo log information to redo log files.
⇒ Writes before the DBWn writes and only confirms that a commit operation has been succeeded when the redo log buffer contents are successfully written to the disk.

⇒ Occurs when:

- Uses commit
- 1MB if redo is there
- Every 3 seconds
- Before DBWn writes
- 1/3rd full

⇒ Manages/Handles damaged redo log files in group configurations.
⇒ Ensures transaction completeness by writing commit records.

5) ARCn:

- Stands for Archiver.
- Copies redo logs to specified locations when a redo log fills up.
- Ensures redo log files are available for future use.
- Up to 10 ARCn processes can be started per instance.
- Prevents performance problems by timely archiving.
- Solutions to performance issues:
 - Increase redo log size
 - Add more redo log groups
 - Increase ARCn processes

⇒ The archive process finishes before the filled redo log is needed again.

6) CKPT:-

- ⇒ Stands for checkpoint process.
- ⇒ Writes info to update the DB control files and headers of datafiles.
- ⇒ Reduces instance recovery time by advancing the checkpoint.
- ⇒ Occurs automatically on redo log file switch.

7) RECO:

- ⇒ Stands for recovery process.
- ⇒ Handles failures of distributed transactions.
- ⇒ Rolls back failed transactions involving changes to tables in multiple databases.

8) MMON & MMNL:

- manageability monitor process

9) MMON:

- Performs tasks related to AWR.

MMNL:-

- ⇒ Manageability monitor for the process - writes statistics from the active session history.

Optional Background Processes:

• ARCn

• CIOO

• Dnn

Diagram:

